Mohammed Abduljabbar

PhD Informatics and Computing

CONTACT

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Google Scholar

TECHNICAL SKILLS

- Deep Learning
- Computer Vision
- Spatiotemporal Modeling
- Transformer Models
- Robotics & Human Aug.
- ML Pipelines / PyTorch

SKILLS & EXPERTISE

Python · C++ · Go · PHP · JavaScript · Full Stack · Linux · Git · Kubernetes · DevOps · Cloud Computing (Azure, AWS, GCP, DigitalOcean) · Edge Computing · IoT · Apache · Nginx · NVIDIA DGX-1 · NVIDIA Jetson · Raspberry Pi · Arduino · 3D Modeling and Animation · Unity Engine · Unreal Engine · Blender · GIS · LLM · VLM · vLLM · Ollama · LangChain · LangGraph · Vector Databases · Hugging Face · LoRA · QLoRA · Unsloth · TensorFlow.js · PyTorch · MediaPipe · Cursor · Claude Code · Gemini · ChatGPT · DeepSeek · Qwen · Blockchain · Automated Crypto Spot / Futures Bots · RAG · Agentic AI · Vibe Coding

LANGUAGES

English	●●●●●
Arabic	●●●●●
Japanese	●○○○○
German	●○○○○

About Me

PhD in Informatics and Computing with deep expertise in deep learning, spatiotemporal modeling, robotics, and deployable AI systems. Research focuses on lightweight architectures for real-world applications — computer vision, Arabic Sign Language recognition using Transformer-based and state-space (Mamba) models, 3D digital twins for human–machine interaction, and Mediapipe-based real-time perception pipelines.

Professional background spans scalable web platforms, cloud-native systems, AI-powered applications, DevOps, robotics systems, and automated trading infrastructure. Passionate about bridging academic research and industry deployment.

What I Do

Deep Learning Research

Designing and training spatiotemporal models, transformers, and state-space architectures for real-world AI.

Computer Vision

Sign language recognition, 3D digital twins, and real-time Mediapipe perception pipelines.

Robotics & Augmentation

ROS-based autonomous mobile robotics (Husky platform), humanoid social robotics (NAO, Pepper), brain–computer interface (BCI) systems, cognitive assistive systems for elderly care, and haptic interfaces for space exploration simulations.

Full Stack Development

Cloud-native platforms, AI-powered web apps, DevOps pipelines, and automated trading systems.

Experience

Oct 2026 — Present
STEM Instructor (Part-Time)
Ada Tech Institute — Dubai Knowledge Park, Dubai, UAE
Project-based STEM coding, robotics, electronics & 3D modelling (Tinkercad) · Python instruction from Turtle to game dev & intro AI · Arduino, Micro:bit & LEGO SPIKE Prime · Curriculum & Python/Colab materials · Portfolio-ready student projects.

Jan 2020 — May 2026
PhD Fellow, Researcher & Full Stack Developer
AI, Computer Vision & Robotics Lab — UAEU
Robot explorer for deep wells · Cognitive robotic training for elderly care · Arabic Sign Language Translation · Fine-tuning video transformers for sign language recognition.

Apr 2022 — Dec 2023
PhD Fellow, Researcher & Full Stack Developer
Interactive Robots & Digital Media Lab — UAEU
AI & robotics for human augmentation and Mars exploration · Astro Haptic Footwear research grant · AI-driven applications and research platforms.

Jan 2024 — Jun 2024
Adjunct Instructor
United Arab Emirates University
Taught Programming Lab I (CSBP121): structured programming, algorithms, arrays, selection & repetition structures.

Sep 2023 — Jan 2024
Higher Education Teaching Instructor
United Arab Emirates University
Completed: Center of Excellence in Teaching and Learning (CETL) PhD Teaching Academy (Cohort 8).

Apr 2022 — Jan 2024
Research Assistant
Zayed University
Astro Haptic Footwear research grant · AI and robotics research development.

Sep–Nov 2023
Research Paper Reviewer
UAEU — IIT'23 Conference
Reviewer for the International Conference on Innovations in Information Technology.

Jan 2018 — Jan 2019
IT Supervisor & Web Developer
Mansion Computer Gaming Center
Supervised IT ops across two branches · Developed company platform and POS system.

Dec 2014 — Mar 2015
Web Developer / Project Coordinator
AsfarTrip — Asfar Group, Dubai, UAE
Web Developer and Software Engineer at Asfar Group · 4 mos.

Dec 2010 — May 2014
Research Developer
KDD Lab — Kansas State University
Web apps and API integration · Python data processing · GIS-related research publications.

Aug 2013 — Dec 2013
Teaching Assistant / Lab Assistant
Kansas State University
TA for Introductory Computer Science course · Scratch programming · Assisted students in labs, exams, and assignments · Graded projects, assignments, and exams.

Jul–Nov 2012
IT Help Desk & BI Analyst
Zain KSA
Reports via Oracle, Teradata & SAP · BMC Remedy ticketing · TDWI Business Intelligence training.

Teaching

Programming Lab I (CSBP121)

United Arab Emirates University · Adjunct Instructor
Jan 2024 — Jun 2024

Teaching Assistant / Lab Assistant

Kansas State University
Aug 2013 — Dec 2013

Research Grants & Funded Projects

PhD Fellow / Researcher conducting interdisciplinary research across grant-funded initiatives, spanning artificial intelligence, computer vision, robotics, human augmentation, and intelligent human–machine systems.

Two-Handed Static and Dynamic Arabic Sign Language Translation System Integration Using Spatiotemporal and Machine Translation Models

UAEU Program for Advanced Research (UPAR) · Apr 2022 — Mar 2024

Role: PhD Fellow / Researcher · **PI:** Dr. Munkhjargal Gochoo · **Award:** AED 399,300

Contributed to AI-based Arabic Sign Language translation research using spatiotemporal deep learning, computer vision, temporal modelling, and machine translation.

Astro Haptic Footwear: Human Augmentation for Mars Exploration

UAEU–Zayed University Joint Research Grant · Jan 2022 — Jan 2024

Role: PhD Fellow / Researcher · **PI:** Dr. Jose Berengueres · **Award:** AED 360,500

Contributed to interdisciplinary R&D in human augmentation, haptic interfaces, machine learning, sensing, robotics, prototype development, experimentation, and research dissemination.

Computer-/Robotic-Based Cognitive Training System for Elderly People in the UAE

UAEU Program for Advanced Research (UPAR) · Jan 2019 — Dec 2022

Role: PhD Fellow / Researcher · **PI:** Dr. Fady Shibata Alnajjar · **Award:** AED 673,000

Contributed to AI-enabled cognitive training and autonomous assessment using humanoid social robotics, machine learning, human–robot interaction, and culturally adapted Arabic interactive systems.

Education

PhD in Informatics and Computing

United Arab Emirates University
Jan 2021 — May 2026 · GPA 3.45/4.00

MSc in Software Engineering

United Arab Emirates University
Aug 2019 — Dec 2020 · GPA 3.39/4.00

Postgraduate Courses in Cloud Computing

Dublin City University
Jan 2015 — Jan 2017

BSc in Computer Information Systems — Minor: Business Admin

Kansas State University
Jun 2008 — Dec 2011 · GPA 3.15/4.00

Achievements

- ★ IEEE SMS BR41N Competition — Brain Computer Interface Hackathon **1st Place**
- ★ ACM Programming Competition **2nd Place**
- ★ Arabic Sign Language AI System Demo — **Abu Dhabi 6G Summit**
- ★ Robot Programmer & Operator — **UAEU Pavilion, EXPO Dubai**
- ★ Astrohaptic Project & Space Suit Demonstrations — **NAJAH Education Expo**

References

MG
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